ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** snake	Energy decreases at each step.
3	**(d)** bees and flowers	Both benefit.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(d)** support all the other levels and hold the most energy	All other organisms depend on producers.
6	**(d)** parasitism	One benefits, one harmed.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(b)** increase	Fewer predators.
9	**(b)** algae and fungi	Mutual benefit.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grass is the producer; deer the primary consumer; tiger the secondary consumer (top carnivore) (1); The tiger level has the least energy (1).	1+1
12	Trees take in CO2 for photosynthesis (1); Carbon is stored in wood (1).	1+1
13	Plants form the base (1); Energy decreases upward (1).	1+1
14	Birds eat poisoned insects (1); Poison passes along the food chain (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Energy decreases at each trophic level (1); A small amount of energy reaches the top level (1); So only a few top carnivores can be supported (1).	1+1+1
16	Fertiliser run-off feeds algae (1); Algae die and decay using oxygen (1); Fish die and water stinks (1).	1+1+1
17	Producing meat uses more plant energy (1); Energy lost at each step (1); Plant diets are efficient (1).	1+1+1
	OR: Pollinators die, less crop yield (1); Predators of pests die (1); Birds starve (1).	1+1+1
18	Roots hold soil (1); Leaf litter adds humus (1); Animals and decomposers return (1).	1+1+1
	OR: Oxygen is used up (1); Aquatic life suffers (1); Waste is broken down but at a cost (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Herbivores increase (1); Overgrazing of grass (1); Soil erosion (1); Water and shelter loss (1); Ecosystem degrades; other species decline (1).	1+1+1+1+1
	OR: Global warming (1); Ice melts and sea level rises (1); Ocean acidification (1); Extreme weather (1); Individual: save energy and plant trees (1).	1+1+1+1+1
20	Chains: grass, grasshopper, frog, snake, hawk (1); Grass, rabbit, hawk; grass, mouse, snake, hawk (1); Hawk: third, fourth and fifth levels (1); Without frogs: grasshoppers rise, snakes switch to mice (1); Network adjusts but balance is changed (1).	1+1+1+1+1
	OR: Pesticide on plants (1); Insects eating them take it in (1); Frogs and snakes eating insects take it in (1); Harm to useful organisms (1); Alternatives: neem sprays, organic farming, biological control (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a decrease in dissolved oxygen after the algae die	Eutrophication.
2	**(d)** removal of dead animals	Scavengers.
3	**(c)** A is true but R is false.	A is true; R is false.
4	**(b)** increase of the predators	Predators lose food.
5	**(d)** keeping their numbers in check	Population control.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** grass to grasshopper to frog	From producers upward.
8	**(d)** two animals fighting for the same food	Compete for resource.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(b)** loss of energy between levels	Pyramid of biomass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pulses fix nitrogen (1); Cereals use the soil nitrogen (1).	1+1
12	Roots hold the soil (1); Leaves reduce the speed of rainfall; soil absorbs water (1).	1+1
13	More producer energy in the tropics (1); More niches and species (1).	1+1
14	Mice and rodents increase (1); More damage to crops (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Forest cut reduces space (1); Prey decreases (1); Fewer tigers (1).	1+1+1
16	Not biodegradable (1); Microbes cannot break them (1); They break into microplastics (1).	1+1+1
17	Grasses and herbs appear (1); Shrubs and small trees grow (1); Forest species return (1).	1+1+1
	OR: Predators eat herbivores (1); Herbivore numbers kept stable (1); Vegetation is preserved (1).	1+1+1
18	Ice melts, polar bears lose habitat (1); Warm water bleaches corals (1); Crops suffer from heat (1).	1+1+1
	OR: Absorb CO2 and dust (1); Release oxygen (1); Reduce heat and stress (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Matter recycles; energy flows one-way (1); Carbon cycles (1); Nitrogen cycles (1); Sunlight enters; heat leaves (1); Decomposers complete the cycle (1).	1+1+1+1+1
	OR: Evaporation, transpiration, condensation, precipitation (1); Deforestation reduces transpiration and increases run-off (1); Urbanisation reduces seepage into the ground (1); Rainwater harvesting recharges groundwater (1); Balance of the cycle is restored with trees and harvesting (1).	1+1+1+1+1
20	Habitat loss (1); Poaching (1); Pollution (1); Invasive species (1); Protected areas, laws and community participation (1).	1+1+1+1+1
	OR: Food from plants and animals (1); Oxygen from trees (1); Water from rivers (1); Medicines from plants (1); Pollution and floods when nature is harmed (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** habitat destruction	Loss of home.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** decomposition and recycling of nutrients	Decomposers.
4	**(a)** groundwater level	Recharges.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Fewer trees, less CO_2 uptake.
6	**(b)** pollination of flowering plants	Fruit and seed set.
7	**(c)** reducing waste by reuse, recycling and composting	Sustainable.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(c)** all consumers eventually die	Energy source lost.
10	**(d)** meet present needs without harming the future	Definition.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fish need oxygen for respiration (1); Gills absorb oxygen from water (1).	1+1
12	Energy decreases up the chain (1); Fish need many plants (1).	1+1
13	They decompose by microbes (1); Cause less pollution (1).	1+1
14	Using resources without exhausting them (1); Rainwater harvesting (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Many species live there (1); Offer food and shelter (1); Need protection (1).	1+1+1
16	They protect Khejri trees (1); They protect blackbuck and chinkara (1); Community values (1).	1+1+1
17	They provide space to move (1); Prevent isolation of populations (1); Reduce human-wildlife conflict (1).	1+1+1
	OR: Plants give oxygen and food (1); Fish give CO_2 (1); Snails and microbes decompose wastes (1).	1+1+1
18	Shows relationships (1); Helps to see which species are key (1); Planning conservation (1).	1+1+1
	OR: Make compost (1); Use as animal feed or biogas (1); Reduces pollution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Project Tiger: government programme (1); Reserves protect habitats (1); Chipko: community movement (1); Villagers hugged trees (1); Both work for conservation (1).	1+1+1+1+1
	OR: Urban heat (1); More dust and CO_2 (1); Water table falls (1); Birds and insects disappear (1); Plant trees, rainwater harvesting, green roofs, ponds, parks (1).	1+1+1+1+1
20	Hydrilla in water with a funnel and test tube in sunlight (1); Gas bubbles collect (1); Glowing splinter rekindles (1); Compost pit heats up with microbial activity (1); Needs air and moisture (1).	1+1+1+1+1
	OR: Meet present needs without harming the future (1); Forest committee protects trees (1); Rainwater harvesting (1); Composting of waste (1); Renewable energy such as biogas (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.